Chemistry • SI measurement and data quality

Empirical Data

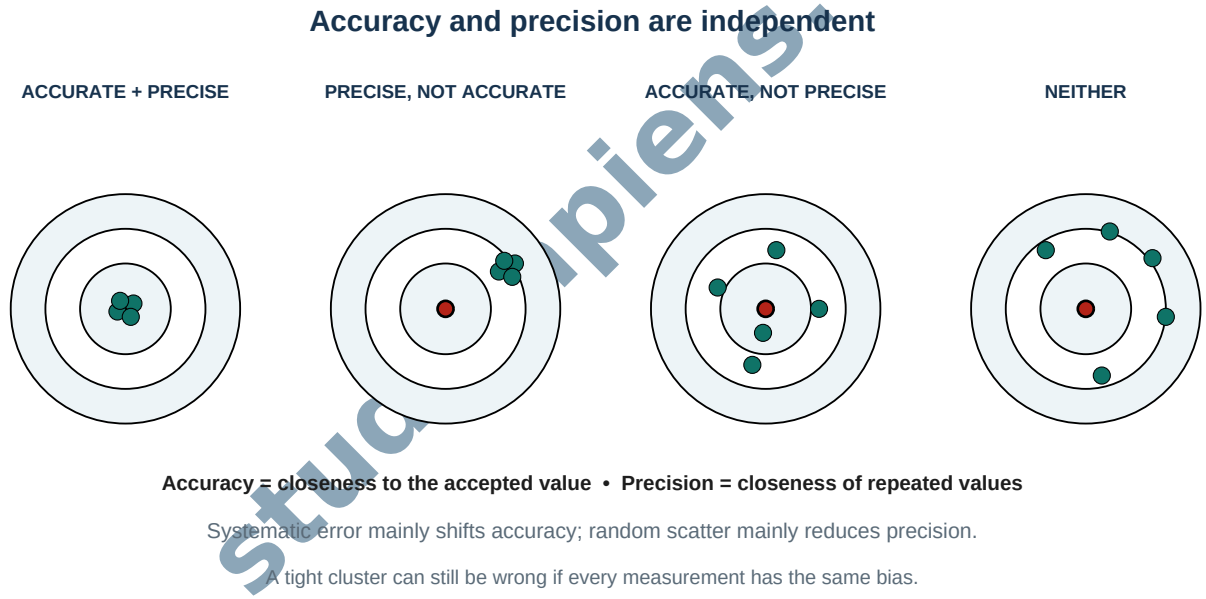
Official exercise • Complete solved answers • Accuracy and precision guide

Chapter 12 • Complete official exercise • 10 MCQs + 8 short + 5 constructed + 5 descriptive + 1 investigative

What this chapter is about

• **SI** provides shared definitions, decimal prefixes and coherent derived units for reproducible measurement.
• The five base quantities emphasized in Chemistry are **length, mass, time, temperature and amount of substance**.
• **Accuracy** concerns closeness to an accepted value; **precision** concerns agreement among repeated measurements.
• **Systematic error** biases results in a consistent direction; **random error** causes unpredictable scatter.

Figure 1 — Original target model: a precise result need not be accurate



Modern SI: The kilogram is defined through the fixed Planck constant, the kelvin through the fixed Boltzmann constant, and the mole contains exactly $6.02214076 \times 10^{23}$ elementary entities. Older prototype-kilogram, water triple-point and approximate-mole descriptions are historical.

A. Multiple Choice Questions

1. Which of the following pairs of quantities may be measured in the same unit?

(a) heat and temperature
(b) temperature and area
(c) heat and work
(d) length and work

Answer: (c) heat and work — Heat and work are both forms of energy transfer and are measured in joules. Temperature is measured in kelvin or °C, area in m² and length in m.

2. In which unit do we usually measure the energy present in food?

(a) kilojoules
(b) megajoules
(c) Calorie
(d) joule

Answer: (c) Calorie — The intended everyday answer is food Calorie. A capital-C food Calorie is 1 kcal = 4184 J; scientific labels also commonly state kJ.

3. What prefix is used for 10^{-12}?

(a) Mega
(b) Pico
(c) Giga
(d) Tessa

Answer: (b) Pico — Pico (p) means 10^{-12}. Mega = 10^{6} and giga = 10^{9}; 'Tessa' is not an SI prefix.

4. In SI, unit of pressure is expressed in:

(a) newton per metre
(b) newton per metre square
(c) joule
(d) pascal

Answer: (b) newton per metre square = (d) pascal — **Both (b) and (d) are correct:** 1 Pa = 1 N m^{-2}. The printed question therefore has two equivalent correct options.

5. Which symbol is used for kilogram in SI units?

(a) K
(b) k
(c) Kgm
(d) kg

Answer: (d) kg — The unit symbol is lower-case kg. K alone denotes kelvin and k alone is the kilo prefix.

6. What does a mole represent?

(a) number
(b) mass
(c) volume
(d) length

Answer: (a) number — The mole is the SI unit of amount of substance and counts exactly $6.02214076 \times 10^{23}$ specified elementary entities.

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7. Which unit of volume should usually be used in Chemistry?

(a) millilitre (b) litre
(c) cubic centimetre (d) cubic metre

Answer: (c) cubic centimetre — The chapter selects cm^3 for manageable laboratory volumes; 1 cm^3 = 1 mL = 10^{-6} m^3. Litres and mL are also accepted practical volume units.

8. Express 0.000840 in scientific notation.

(a) 8.40×10^{-3} (b) 840×10^{-6}
(c) 8.40×10^{-4} (d) 84.0×10^{-5}

Answer: (c) 8.40×10^{-4} — Move the decimal four places right to obtain 8.40; multiply by 10^{-4}. The trailing zero records three significant figures.

9. In SI units, prefix nano means:

(a) 10^{-9} (b) 10^{-8}
(c) 10^{-11} (d) 10^{-12}

Answer: (a) 10^{-9} — Nano (n) = 10^{-9}; pico (p) = 10^{-12}.

10. 65°C is equivalent to:

(a) 208 K (b) 338 K
(c) 403 K (d) 300 K

Answer: (b) 338 K — T/K = θ/°C + 273.15 = 65 + 273.15 = 338.15 K, which rounds to 338 K among the options.

B. Questions for Short Answers

12.1. What is consistency of results?

Consistency means repeated measurements or results agree closely under stated conditions. It is another way of describing good repeatability or precision. Consistent results can still be inaccurate if the same calibration or procedural bias affects every reading.

12.2. Why are SI units user friendly?

SI uses internationally agreed symbols and definitions, decimal prefixes and a coherent set of base/derived units. Conversions therefore use powers of ten, formulas need fewer arbitrary conversion factors, and results can be compared and reproduced across laboratories and countries.

12.3. Define systematic error and random error.

Systematic error is a repeatable bias that shifts measurements in a predictable direction, such as a balance that always reads 0.20 g too high.

Random error is an unpredictable fluctuation that makes repeated readings scatter above and below their mean, such as small timing differences at a stopwatch.

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Systematic error mainly reduces accuracy; random error mainly reduces precision.

12.4. What is the reason behind a random error?

Random errors arise from uncontrolled, unpredictable variations: instrument noise and limited resolution, small temperature or air-current changes, reaction-time differences, slight changes in reading angle or sample handling, and other fluctuations that are not identical in every trial.

12.5. Does systematic error affect accuracy?

Yes. A systematic offset moves the measured mean away from the accepted value, so it directly reduces accuracy. Repeated readings may remain tightly clustered and precise, but averaging them does not remove the bias.

12.6. Which other systems of measurement are used apart from SI units?

Examples include the **CGS** system (centimetre–gram–second), the older **MKS** system (metre–kilogram–second), and **Imperial/US customary** units such as inch, foot, pound and gallon. Local customary units are also used in markets, though their values may vary by place.

12.7. Define metre.

The **metre (m)** is the SI base unit of length. Since 1983 it has been defined by fixing the speed of light in vacuum at exactly 299,792,458 m s^{-1}.

Equivalently, it is the distance travelled by light in vacuum in exactly 1/299,792,458 of a second, with the second defined independently through the caesium-133 transition.

12.8. Mention two benefits scientists get by using SI units.

(1) Clear communication and comparability: the same unit has the same meaning worldwide. **(2) Reliable calculation and replication:** coherent equations and decimal prefixes reduce conversion mistakes, allowing other scientists to check and reproduce experiments. SI also supports safe engineering and data exchange.

C. Constructed Response Questions

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12.1. Compare the units in SI system with those in MKS system.

Feature	MKS	SI
Base quantities	Length, mass and time: m, kg, s	Seven base quantities with units m, kg, s, A, K, mol and cd
Scope	Mainly mechanics	Mechanics plus thermal, electrical, chemical, photometric and other measurements
Derived units	Mechanical combinations such as N, J, Pa	Coherently includes MKS units and extends them to all fields
Status	Historical foundation/subset of SI	Modern internationally agreed measurement system

Thus SI does not contradict MKS; it retains m, kg and s and adds other base units and exact modern definitions.

12.2. What are five basic SI units which are used in Chemistry?

Quantity	SI unit	Symbol	Useful definition / note
Length	metre	m	Defined using the fixed speed of light and the second
Mass	kilogram	kg	Defined using the fixed Planck constant
Time	second	s	9,192,631,770 periods of the specified caesium-133 radiation
Thermodynamic temperature	kelvin	K	Defined using the fixed Boltzmann constant
Amount of substance	mole	mol	Exactly $6.02214076 \times 10^{23}$ elementary entities

Laboratory work commonly uses g, cm³/mL and °C as practical subunits or accepted scales, while calculations may require kg, m³ and K.

12.3. Explain the three units derived from the basic SI units.

Quantity	Definition	Derived SI unit
Area	length × width	m × m = m²
Volume	length × width × height	m × m × m = m³
Density	mass / volume	kg / m³ = kg m^{-3}

Other coherent derived units include force N = kg m s^{-2}, pressure Pa = N m^{-2} and energy J = N m. These are called derived because their definitions combine base units.

12.4. Why do we prefer to use smaller units of mass and volume in Chemistry?

Laboratory samples are often milligrams/grams and microlitres/millilitres rather than kilograms and cubic metres. Smaller units give readable numbers, match molar mass in g mol^{-1} and common glassware graduations, and reduce conversion mistakes. Examples: 0.025 kg = 25 g and 0.000025 m³ = 25 cm³ = 25 mL.

12.5. What difficulties would we encounter if we used different units of measurement in daily life?

• Prices and quantities could be misunderstood in trade, recipes, fuel sales and construction.

• Medical doses, chemical concentrations and machine settings could become unsafe.

• Extra conversions would waste time and introduce arithmetic or factor errors.

• Parts made in different places might not fit, and experiments would be harder to compare or reproduce.

• Units with regional meanings—such as gallon or local maund—could create further ambiguity.

A shared standard prevents these problems, provided the unit and conversion are recorded explicitly.

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D. Descriptive Questions

12.1. What are our indigenous units of measurement of mass, volume and length?

Quantity	Examples from Pakistan / the subcontinent	Approximate or cautionary relation
Mass	ratti, masha, tola, chhatank, seer, maund	1 tola ≈ 11.664 g; maund and seer values varied regionally
Length	ungli, girah, hath, gaz, karam, kos	1 gaz ≈ 0.9144 m in a common modern conversion; other units varied
Capacity / volume	pao, chhatank, seer and household vessels used as market or kitchen measures	Often measured commodity capacity rather than a fixed volume; calibrate before using in cm^3 or L

Important: These customary measures were not universally standardized. A “seer” of grain, a market capacity measure and a modern litre are not interchangeable without a stated local conversion. Scientific records should therefore use SI or clearly defined accepted units.

12.2. Elaborate the difference between precision and accuracy.

Feature	Accuracy	Precision
Meaning	Closeness of a result or mean to the accepted value	Closeness of repeated measurements to one another
Main error indicator	Systematic bias	Random scatter / standard deviation
Improve by	Calibration, correction and validated method	Repetition, controlled conditions and better resolution
Can exist without the other?	Yes: a widely scattered set may average near truth	Yes: a tight cluster may have a constant offset

Example with accepted mass 20.0 g: 17.2, 17.3 and 17.4 g are precise but not accurate. 19.8, 20.5 and 19.6 g have mean ≈19.97 g and are more accurate on average but less precise. Ideally results are both accurate and precise.

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12.3. How can you avoid systematic errors in your measurements?

1. Calibrate balances, pipettes, thermometers and other instruments against traceable standards.

2. Check zero and apply corrections only after determining their sign and magnitude; record the correction.

3. Use the correct validated method: read meniscus at eye level, avoid parallax, rinse/condition glassware appropriately and use the specified temperature.

4. Control environmental factors such as temperature, vibration, drafts and contamination.

5. Use blanks, controls and reference materials to detect background or method bias.

6. Cross-check with a second calibrated instrument or independent method when consequences are important.

Repetition and averaging reduce random scatter but do **not** eliminate a systematic error.

12.4. How do measurements in SI units ensure safety and reliability?

Common definitions let every person interpret the same mass, dose, pressure, volume, temperature or concentration correctly. Coherent formulas reduce hidden conversion factors, while prefixes provide clear scales from nano- to mega-. This supports correct instrument design, chemical preparation, quality control and replication.

For example, writing 250 mg rather than an ambiguous “quarter” dose, or pressure in kPa with an instrument range, makes checks possible. SI improves safety only when users also preserve significant figures, record units at every step, verify conversions and avoid confusing similar symbols such as m/M, mg/Mg and cm^3/m^3.

12.5. Can a student be both inaccurate and imprecise in measurements?

Yes. Inaccuracy means the mean is far from the accepted value; imprecision means the individual readings are widely scattered.

Accepted mass = 20.0 g

Readings = 17.1 g, 19.4 g, 16.8 g

Mean = (17.1 + 19.4 + 16.8)/3 = 17.77 g

Large offset from 20.0 g + large scatter = inaccurate and imprecise

Possible causes are an uncalibrated balance (systematic bias) together with unstable handling or environmental fluctuations (random error). Calibration and repeated controlled measurements address different parts of the problem.

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E. Investigative Question

12.1. Elaborate the importance of using SI units in space exploration.

Spacecraft navigation combines distance, velocity, time, mass, force and energy over enormous distances and long missions. A small unit error can grow into a major trajectory error. SI gives international teams one coherent language for design, simulation, launch, communication and docking.

• **Propulsion:** thrust in N and impulse in N s must match navigation equations using kg, m and s.

• **Trajectory:** consistent km/m and s values are essential for orbital timing, course corrections and rendezvous.

• **Thermal and life-support systems:** K, Pa, mol and energy units regulate safe cabin and fuel conditions.

• **International hardware and data:** common units let separately built components and scientific observations be integrated and checked.

Historical warning: NASA’s Mars Climate Orbiter was lost in 1999 after a team supplied thrust data in pound-force seconds while navigation software expected newton-seconds. The failure illustrates why unit labels, interface checks and dimensional analysis are mission-critical.

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Points to Remember

Five SI base units emphasized: metre (m), kilogram (kg), second (s), kelvin (K), mole (mol).

1 Pa = 1 N m^{-2}, so MCQ (iv) has two correct printed choices: (b) and (d).

1 cm^3 = 1 mL = 10^{-6} m^3; use g and cm^3/mL for convenient laboratory-scale values.

T/K = θ/°C + 273.15; 65°C = 338.15 K ≈ 338 K.

pico = 10^{-12}; nano = 10^{-9}; 0.000840 = 8.40 × 10^{-4} (3 significant figures).

Accuracy ≠ precision: calibration addresses bias; repetition and averaging address random scatter.

Modern SI units use fixed physical constants, not a prototype kilogram or the triple-point definition of kelvin.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	c	2	c	3	b	4	b/d	5	d
6	a	7	c	8	c	9	a	10	b

How to attempt this chapter in the paper

• Write the unit after every numerical value and check dimensions before accepting a result.
• For prefix conversions, state the power of ten first; for scientific notation use one non-zero digit before the decimal.
• For error questions, name the error type, its effect on accuracy/precision and a matching remedy.
• Do not assume a customary unit has one universal value; provide a location and verified conversion if used.